Orbital-shadow covariance coefficients

$k = 1$**, class I,** $(1, 2; 2, 1)$ Class I: impossible charge, so no coefficients are defined.

$k = 1$**, class II,** $(1, 1; 1, 1)$ $L_+ = \emptyset$, $L_- = \emptyset$, $L_0 = \{1\}$.

$$A_\emptyset = \frac{\eta\,(N - \eta + 1)}{N + 2}$$
$$A_{\{1\}} = \frac{(\eta + 2)\,(N - \eta + 1)}{N + 2}$$

$$\Gamma_\emptyset = \frac{\eta\,(N - \eta + 1)}{N + 2}$$
$$\Gamma_{\{1\}} = \frac{2\,(N - \eta + 1)}{N + 2}$$

$k = 1$**, class III,** $(1, 1; 2, 2)$ $L_+ = \emptyset$, $L_- = \emptyset$, $L_0 = \{1, 2\}$.

$$A_\emptyset = -\frac{\eta\,(N - \eta + 1)}{N\,(N + 2)}$$
$$A_{\{1\}} = -\frac{(\eta + 1)\,(N - \eta + 1)}{N\,(N + 2)}$$
$$A_{\{2\}} = -\frac{(\eta + 1)\,(N - \eta + 1)}{N\,(N + 2)}$$
$$A_{\{1,2\}} = \frac{N^2 - N\eta + N + \eta^2 + \eta}{N\,(N + 2)}$$

$$\Gamma_\emptyset = -\frac{\eta\,(N - \eta + 1)}{N\,(N + 2)}$$
$$\Gamma_{\{1\}} = -\frac{N - \eta + 1}{N\,(N + 2)}$$
$$\Gamma_{\{2\}} = -\frac{N - \eta + 1}{N\,(N + 2)}$$
$$\Gamma_{\{1,2\}} = \frac{N + 1}{N}$$

$k = 1$**, class III',** $(1, 2; 1, 2)$ $L_+ = \emptyset$, $L_- = \emptyset$, $L_0 = \{1, 2\}$.

$$A_\emptyset = \frac{\eta\,(N + 1)\,(N - \eta + 1)}{N\,(N + 2)}$$
$$A_{\{1\}} = \frac{(N + 1)\,(\eta + 1)\,(N - \eta + 1)}{N\,(N + 2)}$$
$$A_{\{2\}} = \frac{(N + 1)\,(\eta + 1)\,(N - \eta + 1)}{N\,(N + 2)}$$
$$A_{\{1,2\}} = \frac{(N + 1)\,(N - \eta)\,(\eta + 1)}{N\,(N + 2)}$$

$$\Gamma_{\emptyset} = \frac{\eta\,(N+1)\,(N-\eta+1)}{N\,(N+2)}$$
$$\Gamma_{\{1\}} = \frac{(N+1)\,(N-\eta+1)}{N\,(N+2)}$$
$$\Gamma_{\{2\}} = \frac{(N+1)\,(N-\eta+1)}{N\,(N+2)}$$
$$\Gamma_{\{1,2\}} = -\frac{N+1}{N}$$

$k=1$**, class IV,** $(1,2;1,1)$ $\quad L_+ = \{1\}$, $L_- = \{2\}$, $L_0 = \emptyset$.

$$A_{\emptyset} = \frac{N-\eta+1}{N+2}$$

$$\Gamma_{\emptyset} = \frac{N-\eta+1}{N+2}$$

$k=1$**, class V,** $(1,2;1,3)$ $\quad L_+ = \{3\}$, $L_- = \{2\}$, $L_0 = \{1\}$.

$$A_{\emptyset} = \frac{(N+1)\,(N-\eta+1)}{N\,(N+2)}$$
$$A_{\{1\}} = -\frac{(N+1)\,(\eta+1)}{N\,(N+2)}$$

$$\Gamma_{\emptyset} = \frac{(N+1)\,(N-\eta+1)}{N\,(N+2)}$$
$$\Gamma_{\{1\}} = -\frac{N+1}{N}$$

$k=1$**, class V',** $(1,1;2,3)$ $\quad L_+ = \{3\}$, $L_- = \{2\}$, $L_0 = \{1\}$.

$$A_{\emptyset} = -\frac{N-\eta+1}{N\,(N+2)}$$
$$A_{\{1\}} = \frac{N^2+2N+\eta+1}{N\,(N+2)}$$

$$\Gamma_{\emptyset} = -\frac{N-\eta+1}{N\,(N+2)}$$
$$\Gamma_{\{1\}} = \frac{N+1}{N}$$

$k=1$**, class VI,** $(1,2;3,4)$ $\quad L_+ = \{1,4\}$, $L_- = \{2,3\}$, $L_0 = \emptyset$.

$$A_{\emptyset} = \frac{N+1}{N}$$

$$\Gamma_{\emptyset} = \frac{N+1}{N}$$

$k = 2$**, class I,** $(12, 34; 56, 17)$ Class I: impossible charge, so no coefficients are defined.

$k = 2$**, class II,** $(12, 12; 12, 12)$ $L_+ = \emptyset$, $L_- = \emptyset$, $L_0 = \{1, 2\}$.

$$
\begin{aligned}
A_{\emptyset} &= \frac{\eta\,(\eta-1)\,(N-\eta+1)^2}{2N\,(N+2)} \\
A_{\{1\}} &= \frac{(\eta-1)\,(N-\eta+1)\left(N\eta+2N-\eta^2+1\right)}{2N\,(N+2)} \\
A_{\{2\}} &= \frac{(\eta-1)\,(N-\eta+1)\left(N\eta+2N-\eta^2+1\right)}{2N\,(N+2)} \\
A_{\{1,2\}} &= \frac{(N-\eta+1)\left(N\eta^2+2N\eta+3N-\eta^3+\eta\right)}{2N\,(N+2)}
\end{aligned}
$$

$$
\begin{aligned}
\Gamma_{\emptyset} &= \frac{\eta\,(\eta-1)\,(N-\eta+1)^2}{2N\,(N+2)} \\
\Gamma_{\{1\}} &= \frac{(\eta-1)\,(N-\eta+1)\,(2N-\eta+1)}{2N\,(N+2)} \\
\Gamma_{\{2\}} &= \frac{(\eta-1)\,(N-\eta+1)\,(2N-\eta+1)}{2N\,(N+2)} \\
\Gamma_{\{1,2\}} &= -\frac{(N-\eta+1)\,(N\eta-7N+2\eta-2)}{2N\,(N+2)}
\end{aligned}
$$

$k = 2$**, class III,** $(12, 12; 13, 13)$ $L_+ = \emptyset$, $L_- = \emptyset$, $L_0 = \{1, 2, 3\}$.

$$
\begin{aligned}
A_{\emptyset} &= -\frac{\eta\,(\eta-1)\,(N-\eta+1)\,(2N-3\eta+3)}{4N\,(N-1)\,(N+2)} \\
A_{\{1\}} &= -\frac{(\eta-1)\,(N-\eta+1)\left(2N\eta+4N-3\eta^2+\eta+2\right)}{4N\,(N-1)\,(N+2)} \\
A_{\{2\}} &= -\frac{(\eta-1)\,(N-\eta+1)\left(N\eta+3N-3\eta^2+\eta+2\right)}{4N\,(N-1)\,(N+2)} \\
A_{\{3\}} &= -\frac{(\eta-1)\,(N-\eta+1)\left(N\eta+3N-3\eta^2+\eta+2\right)}{4N\,(N-1)\,(N+2)} \\
A_{\{1,2\}} &= -\frac{(\eta+1)\,(N-\eta+1)\left(N\eta+2N-3\eta^2+5\eta-2\right)}{4N\,(N-1)\,(N+2)} \\
A_{\{1,3\}} &= -\frac{(\eta+1)\,(N-\eta+1)\left(N\eta+2N-3\eta^2+5\eta-2\right)}{4N\,(N-1)\,(N+2)} \\
A_{\{2,3\}} &= \frac{(N-\eta+1)\left(N^2\eta-3N^2-3N\eta+N+3\eta^3-2\eta^2-3\eta+2\right)}{4N\,(N-1)\,(N+2)} \\
A_{\{1,2,3\}} &= \frac{(N-\eta+1)\left(N^2\eta+5N^2-N\eta-5N+3\eta^3-3\eta\right)}{4N\,(N-1)\,(N+2)}
\end{aligned}
$$

$$
\Gamma_{\emptyset} = -\frac{\eta\,(\eta-1)\,(N-\eta+1)\,(2N-3\eta+3)}{4N\,(N-1)\,(N+2)}
$$

$$\Gamma_{\{1\}} = -\frac{(\eta-1)(N-\eta+1)(2N-\eta+1)}{2N(N-1)(N+2)}$$
$$\Gamma_{\{2\}} = \frac{(\eta-1)(N-\eta+1)(N\eta-3N+2\eta-2)}{4N(N-1)(N+2)}$$
$$\Gamma_{\{3\}} = \frac{(\eta-1)(N-\eta+1)(N\eta-3N+2\eta-2)}{4N(N-1)(N+2)}$$
$$\Gamma_{\{1,2\}} = \frac{(N-\eta+1)(3N\eta-9N+2\eta-2)}{4N(N-1)(N+2)}$$
$$\Gamma_{\{1,3\}} = \frac{(N-\eta+1)(3N\eta-9N+2\eta-2)}{4N(N-1)(N+2)}$$
$$\Gamma_{\{2,3\}} = \frac{(N+1)(N-\eta+1)(N\eta-3N+2\eta-2)}{4N(N-1)(N+2)}$$
$$\Gamma_{\{1,2,3\}} = \frac{2(N+1)(N-\eta+1)}{(N-1)(N+2)}$$

$k=2$**, class III',** $(12,13;12,13)$ $\quad L_+=\emptyset$, $L_-=\emptyset$, $L_0=\{1,2,3\}$.

$$A_{\emptyset} = \frac{\eta(\eta-1)(N-\eta+1)\left(2N^2-2N\eta+2N-\eta+1\right)}{4N(N-1)(N+2)}$$
$$A_{\{1\}} = \frac{(\eta-1)(N-\eta+1)\left(2N^2\eta+4N^2-2N\eta^2+2N-\eta^2+\eta\right)}{4N(N-1)(N+2)}$$
$$A_{\{2\}} = \frac{(\eta-1)(\eta+1)(N-\eta+1)\left(2N^2-2N\eta+2N-\eta+1\right)}{4N(N-1)(N+2)}$$
$$A_{\{3\}} = \frac{(\eta-1)(\eta+1)(N-\eta+1)\left(2N^2-2N\eta+2N-\eta+1\right)}{4N(N-1)(N+2)}$$
$$A_{\{1,2\}} = \frac{(\eta+1)(N-\eta+1)^2\cdot(2N\eta+N+\eta-1)}{4N(N-1)(N+2)}$$
$$A_{\{1,3\}} = \frac{(\eta+1)(N-\eta+1)^2\cdot(2N\eta+N+\eta-1)}{4N(N-1)(N+2)}$$
$$A_{\{2,3\}} = \frac{(\eta+1)(N-\eta+1)\left(2N^2\eta-N^2-2N\eta^2+2N\eta+N-\eta^2+\eta\right)}{4N(N-1)(N+2)}$$
$$A_{\{1,2,3\}} = \frac{(N-\eta)(\eta+1)(N-\eta+1)(2N\eta+N+\eta-1)}{4N(N-1)(N+2)}$$

$$\Gamma_{\emptyset} = \frac{\eta(\eta-1)(N-\eta+1)\left(2N^2-2N\eta+2N-\eta+1\right)}{4N(N-1)(N+2)}$$
$$\Gamma_{\{1\}} = \frac{(\eta-1)(N-\eta+1)(2N-\eta+1)}{2(N-1)(N+2)}$$
$$\Gamma_{\{2\}} = \frac{(\eta-1)(N-\eta+1)\left(2N^2-2N\eta+2N-\eta+1\right)}{4N(N-1)(N+2)}$$
$$\Gamma_{\{3\}} = \frac{(\eta-1)(N-\eta+1)\left(2N^2-2N\eta+2N-\eta+1\right)}{4N(N-1)(N+2)}$$
$$\Gamma_{\{1,2\}} = -\frac{(N-\eta+1)(N\eta-7N+4\eta-4)}{4(N-1)(N+2)}$$

$$\Gamma_{\{1,3\}} = -\frac{(N-\eta+1)(N\eta-7N+4\eta-4)}{4(N-1)(N+2)}$$
$$\Gamma_{\{2,3\}} = -\frac{(N+1)(N-\eta+1)(N\eta-3N+2\eta-2)}{4N(N-1)(N+2)}$$
$$\Gamma_{\{1,2,3\}} = -\frac{2(N+1)(N-\eta+1)}{(N-1)(N+2)}$$

$k=2$**, class IV,** $(12,12;34,34)\quad L_+=\emptyset,\ L_-=\emptyset,\ L_0=\{1,2,3,4\}.$

$$A_{\emptyset} = \frac{\eta(\eta-1)(N-2\eta+2)(N-\eta+1)}{N(N-2)(N-1)(N+2)}$$
$$A_{\{1\}} = \frac{(\eta-1)(N-\eta+1)\left(N\eta+3N-4\eta^2+2\eta+2\right)}{2N(N-2)(N-1)(N+2)}$$
$$A_{\{2\}} = \frac{(\eta-1)(N-\eta+1)\left(N\eta+3N-4\eta^2+2\eta+2\right)}{2N(N-2)(N-1)(N+2)}$$
$$A_{\{3\}} = \frac{(\eta-1)(N-\eta+1)\left(N\eta+3N-4\eta^2+2\eta+2\right)}{2N(N-2)(N-1)(N+2)}$$
$$A_{\{4\}} = \frac{(\eta-1)(N-\eta+1)\left(N\eta+3N-4\eta^2+2\eta+2\right)}{2N(N-2)(N-1)(N+2)}$$
$$A_{\{1,2\}} = \frac{(\eta+1)(N-\eta+1)\left(N-2\eta^2+4\eta-2\right)}{N(N-2)(N-1)(N+2)}$$
$$A_{\{1,3\}} = -\frac{(N-\eta+1)\left(N^2\eta-3N^2-8N\eta+4N+8\eta^3-8\eta^2-4\eta+4\right)}{4N(N-2)(N-1)(N+2)}$$
$$A_{\{1,4\}} = -\frac{(N-\eta+1)\left(N^2\eta-3N^2-8N\eta+4N+8\eta^3-8\eta^2-4\eta+4\right)}{4N(N-2)(N-1)(N+2)}$$
$$A_{\{2,3\}} = -\frac{(N-\eta+1)\left(N^2\eta-3N^2-8N\eta+4N+8\eta^3-8\eta^2-4\eta+4\right)}{4N(N-2)(N-1)(N+2)}$$
$$A_{\{2,4\}} = -\frac{(N-\eta+1)\left(N^2\eta-3N^2-8N\eta+4N+8\eta^3-8\eta^2-4\eta+4\right)}{4N(N-2)(N-1)(N+2)}$$
$$A_{\{3,4\}} = \frac{(\eta+1)(N-\eta+1)\left(N-2\eta^2+4\eta-2\right)}{N(N-2)(N-1)(N+2)}$$
$$A_{\{1,2,3\}} = -\frac{(\eta+1)(N-\eta+1)\left(N^2+N\eta-3N+4\eta^2-6\eta+2\right)}{2N(N-2)(N-1)(N+2)}$$
$$A_{\{1,2,4\}} = -\frac{(\eta+1)(N-\eta+1)\left(N^2+N\eta-3N+4\eta^2-6\eta+2\right)}{2N(N-2)(N-1)(N+2)}$$
$$A_{\{1,3,4\}} = -\frac{(\eta+1)(N-\eta+1)\left(N^2+N\eta-3N+4\eta^2-6\eta+2\right)}{2N(N-2)(N-1)(N+2)}$$
$$A_{\{2,3,4\}} = -\frac{(\eta+1)(N-\eta+1)\left(N^2+N\eta-3N+4\eta^2-6\eta+2\right)}{2N(N-2)(N-1)(N+2)}$$
$$A_{\{1,2,3,4\}} = \frac{N^4-N^3\eta-2N^3+3N^2\eta-N^2-N\eta^3-N\eta+2N+2\eta^4-2\eta^3-2\eta^2+2\eta}{N(N-2)(N-1)(N+2)}$$

$$\Gamma_{\emptyset} = \frac{\eta(\eta-1)(N-2\eta+2)(N-\eta+1)}{N(N-2)(N-1)(N+2)}$$

$$
\begin{aligned}
\Gamma_{\{1\}} &= -\frac{(\eta-1)(N-\eta+1)(N\eta-3N+2\eta-2)}{2N(N-2)(N-1)(N+2)} \\
\Gamma_{\{2\}} &= -\frac{(\eta-1)(N-\eta+1)(N\eta-3N+2\eta-2)}{2N(N-2)(N-1)(N+2)} \\
\Gamma_{\{3\}} &= -\frac{(\eta-1)(N-\eta+1)(N\eta-3N+2\eta-2)}{2N(N-2)(N-1)(N+2)} \\
\Gamma_{\{4\}} &= -\frac{(\eta-1)(N-\eta+1)(N\eta-3N+2\eta-2)}{2N(N-2)(N-1)(N+2)} \\
\Gamma_{\{1,2\}} &= -\frac{2(\eta-2)(N-\eta+1)}{(N-2)(N-1)(N+2)} \\
\Gamma_{\{1,3\}} &= -\frac{(N-\eta+1)(N\eta-3N+2\eta-2)}{4N(N-2)(N-1)} \\
\Gamma_{\{1,4\}} &= -\frac{(N-\eta+1)(N\eta-3N+2\eta-2)}{4N(N-2)(N-1)} \\
\Gamma_{\{2,3\}} &= -\frac{(N-\eta+1)(N\eta-3N+2\eta-2)}{4N(N-2)(N-1)} \\
\Gamma_{\{2,4\}} &= -\frac{(N-\eta+1)(N\eta-3N+2\eta-2)}{4N(N-2)(N-1)} \\
\Gamma_{\{3,4\}} &= -\frac{2(\eta-2)(N-\eta+1)}{(N-2)(N-1)(N+2)} \\
\Gamma_{\{1,2,3\}} &= -\frac{2(N+1)(N-\eta+1)}{(N-2)(N-1)(N+2)} \\
\Gamma_{\{1,2,4\}} &= -\frac{2(N+1)(N-\eta+1)}{(N-2)(N-1)(N+2)} \\
\Gamma_{\{1,3,4\}} &= -\frac{2(N+1)(N-\eta+1)}{(N-2)(N-1)(N+2)} \\
\Gamma_{\{2,3,4\}} &= -\frac{2(N+1)(N-\eta+1)}{(N-2)(N-1)(N+2)} \\
\Gamma_{\{1,2,3,4\}} &= \frac{N(N+1)}{(N-2)(N-1)}
\end{aligned}
$$

$k=2$, class IV', $(12,13;24,34)\quad L_+=\emptyset,\ L_-=\emptyset,\ L_0=\{1,2,3,4\}$.

$$
\begin{aligned}
A_{\emptyset} &= -\frac{\eta(\eta-1)(N-\eta+1)\left(2N^2-3N\eta+3N-2\eta+2\right)}{4N(N-2)(N-1)(N+2)} \\
A_{\{1\}} &= -\frac{(\eta-1)(N-\eta+1)\left(N^2\eta+3N^2-3N\eta^2+N\eta+2N-2\eta^2+2\eta\right)}{4N(N-2)(N-1)(N+2)} \\
A_{\{2\}} &= -\frac{(\eta-1)(\eta+1)(N-\eta+1)\left(2N^2-3N\eta+3N-2\eta+2\right)}{4N(N-2)(N-1)(N+2)} \\
A_{\{3\}} &= -\frac{(\eta-1)(\eta+1)(N-\eta+1)\left(2N^2-3N\eta+3N-2\eta+2\right)}{4N(N-2)(N-1)(N+2)} \\
A_{\{4\}} &= -\frac{(\eta-1)(N-\eta+1)\left(N^2\eta+3N^2-3N\eta^2+N\eta+2N-2\eta^2+2\eta\right)}{4N(N-2)(N-1)(N+2)}
\end{aligned}
$$

$$A_{\{1,2\}} = -\frac{(\eta+1)(N-\eta+1)\left(N^2\eta+N^2-3N\eta^2+4N\eta-N-2\eta^2+4\eta-2\right)}{4N(N-2)(N-1)(N+2)}$$
$$A_{\{1,3\}} = -\frac{(\eta+1)(N-\eta+1)\left(N^2\eta+N^2-3N\eta^2+4N\eta-N-2\eta^2+4\eta-2\right)}{4N(N-2)(N-1)(N+2)}$$
$$A_{\{1,4\}} = \frac{(N-\eta+1)\left(N^3\eta-3N^3-4N^2\eta+4N^2+3N\eta^3-2N\eta^2-5N\eta+4N+2\eta^3-4\eta^2+2\eta\right)}{4N(N-2)(N-1)(N+2)}$$
$$A_{\{2,3\}} = -\frac{(\eta+1)(N-\eta+1)\left(2N^2\eta-N^2-3N\eta^2+3N\eta+2N-2\eta^2+2\eta\right)}{4N(N-2)(N-1)(N+2)}$$
$$A_{\{2,4\}} = -\frac{(\eta+1)(N-\eta+1)\left(N^2\eta+N^2-3N\eta^2+4N\eta-N-2\eta^2+4\eta-2\right)}{4N(N-2)(N-1)(N+2)}$$
$$A_{\{3,4\}} = -\frac{(\eta+1)(N-\eta+1)\left(N^2\eta+N^2-3N\eta^2+4N\eta-N-2\eta^2+4\eta-2\right)}{4N(N-2)(N-1)(N+2)}$$
$$A_{\{1,2,3\}} = -\frac{\eta(\eta+1)(N-\eta+1)\left(N^2-3N\eta+N-2\eta+2\right)}{4N(N-2)(N-1)(N+2)}$$
$$A_{\{1,2,4\}} = \frac{(\eta+1)(N-\eta+1)\left(N^3-2N^2+3N\eta^2-2N\eta-N+2\eta^2-4\eta+2\right)}{4N(N-2)(N-1)(N+2)}$$
$$A_{\{1,3,4\}} = \frac{(\eta+1)(N-\eta+1)\left(N^3-2N^2+3N\eta^2-2N\eta-N+2\eta^2-4\eta+2\right)}{4N(N-2)(N-1)(N+2)}$$
$$A_{\{2,3,4\}} = -\frac{\eta(\eta+1)(N-\eta+1)\left(N^2-3N\eta+N-2\eta+2\right)}{4N(N-2)(N-1)(N+2)}$$
$$A_{\{1,2,3,4\}} = \frac{(N-\eta)(\eta+1)\left(N^3-2N^2+3N\eta^2-2N\eta-N+2\eta^2-4\eta+2\right)}{4N(N-2)(N-1)(N+2)}$$

$$\Gamma_{\emptyset} = -\frac{\eta(\eta-1)(N-\eta+1)\left(2N^2-3N\eta+3N-2\eta+2\right)}{4N(N-2)(N-1)(N+2)}$$
$$\Gamma_{\{1\}} = \frac{(\eta-1)(N-\eta+1)(N\eta-3N+2\eta-2)}{4(N-2)(N-1)(N+2)}$$
$$\Gamma_{\{2\}} = -\frac{(\eta-1)(N-\eta+1)\left(2N^2-3N\eta+3N-2\eta+2\right)}{4N(N-2)(N-1)(N+2)}$$
$$\Gamma_{\{3\}} = -\frac{(\eta-1)(N-\eta+1)\left(2N^2-3N\eta+3N-2\eta+2\right)}{4N(N-2)(N-1)(N+2)}$$
$$\Gamma_{\{4\}} = \frac{(\eta-1)(N-\eta+1)(N\eta-3N+2\eta-2)}{4(N-2)(N-1)(N+2)}$$
$$\Gamma_{\{1,2\}} = \frac{(N-\eta+1)(N\eta-3N+2\eta-2)}{2(N-2)(N-1)(N+2)}$$
$$\Gamma_{\{1,3\}} = \frac{(N-\eta+1)(N\eta-3N+2\eta-2)}{2(N-2)(N-1)(N+2)}$$
$$\Gamma_{\{1,4\}} = \frac{N(N-\eta+1)(N\eta-3N+2\eta-2)}{4(N-2)(N-1)(N+2)}$$
$$\Gamma_{\{2,3\}} = \frac{(N-\eta+1)(N\eta-3N+2\eta-2)}{4N(N-2)(N-1)}$$
$$\Gamma_{\{2,4\}} = \frac{(N-\eta+1)(N\eta-3N+2\eta-2)}{2(N-2)(N-1)(N+2)}$$

$$\Gamma_{\{3,4\}} = \frac{(N-\eta+1)(N\eta-3N+2\eta-2)}{2(N-2)(N-1)(N+2)}$$
$$\Gamma_{\{1,2,3\}} = \frac{2(N+1)(N-\eta+1)}{(N-2)(N-1)(N+2)}$$
$$\Gamma_{\{1,2,4\}} = \frac{N(N+1)(N-\eta+1)}{(N-2)(N-1)(N+2)}$$
$$\Gamma_{\{1,3,4\}} = \frac{N(N+1)(N-\eta+1)}{(N-2)(N-1)(N+2)}$$
$$\Gamma_{\{2,3,4\}} = \frac{2(N+1)(N-\eta+1)}{(N-2)(N-1)(N+2)}$$
$$\Gamma_{\{1,2,3,4\}} = -\frac{N(N+1)}{(N-2)(N-1)}$$

$k=2$**, class IV",** $(12,34;12,34)$ $\quad L_+=\emptyset$, $L_-=\emptyset$, $L_0=\{1,2,3,4\}$.

$$A_{\emptyset} = \frac{N\eta(N-\eta)(\eta-1)(N-\eta+1)}{2(N-2)(N-1)(N+2)}$$
$$A_{\{1\}} = \frac{N(N-\eta)(\eta-1)(\eta+1)(N-\eta+1)}{2(N-2)(N-1)(N+2)}$$
$$A_{\{2\}} = \frac{N(N-\eta)(\eta-1)(\eta+1)(N-\eta+1)}{2(N-2)(N-1)(N+2)}$$
$$A_{\{3\}} = \frac{N(N-\eta)(\eta-1)(\eta+1)(N-\eta+1)}{2(N-2)(N-1)(N+2)}$$
$$A_{\{4\}} = \frac{N(N-\eta)(\eta-1)(\eta+1)(N-\eta+1)}{2(N-2)(N-1)(N+2)}$$
$$A_{\{1,2\}} = \frac{N\eta(N-\eta)(\eta+1)(N-\eta+1)}{2(N-2)(N-1)(N+2)}$$
$$A_{\{1,3\}} = \frac{N(\eta+1)(N-\eta+1)\left(2N\eta-N-2\eta^2+2\right)}{4(N-2)(N-1)(N+2)}$$
$$A_{\{1,4\}} = \frac{N(\eta+1)(N-\eta+1)\left(2N\eta-N-2\eta^2+2\right)}{4(N-2)(N-1)(N+2)}$$
$$A_{\{2,3\}} = \frac{N(\eta+1)(N-\eta+1)\left(2N\eta-N-2\eta^2+2\right)}{4(N-2)(N-1)(N+2)}$$
$$A_{\{2,4\}} = \frac{N(\eta+1)(N-\eta+1)\left(2N\eta-N-2\eta^2+2\right)}{4(N-2)(N-1)(N+2)}$$
$$A_{\{3,4\}} = \frac{N\eta(N-\eta)(\eta+1)(N-\eta+1)}{2(N-2)(N-1)(N+2)}$$
$$A_{\{1,2,3\}} = \frac{N\eta(\eta+1)(N-\eta-1)(N-\eta+1)}{2(N-2)(N-1)(N+2)}$$
$$A_{\{1,2,4\}} = \frac{N\eta(\eta+1)(N-\eta-1)(N-\eta+1)}{2(N-2)(N-1)(N+2)}$$
$$A_{\{1,3,4\}} = \frac{N\eta(\eta+1)(N-\eta-1)(N-\eta+1)}{2(N-2)(N-1)(N+2)}$$

$$A_{\{2,3,4\}} = \frac{N\eta\,(\eta+1)\,(N-\eta-1)\,(N-\eta+1)}{2\,(N-2)\,(N-1)\,(N+2)}$$
$$A_{\{1,2,3,4\}} = \frac{N\eta\,(N-\eta)\,(\eta+1)\,(N-\eta-1)}{2\,(N-2)\,(N-1)\,(N+2)}$$

$$\Gamma_{\emptyset} = \frac{N\eta\,(N-\eta)\,(\eta-1)\,(N-\eta+1)}{2\,(N-2)\,(N-1)\,(N+2)}$$
$$\Gamma_{\{1\}} = \frac{N\,(N-\eta)\,(\eta-1)\,(N-\eta+1)}{2\,(N-2)\,(N-1)\,(N+2)}$$
$$\Gamma_{\{2\}} = \frac{N\,(N-\eta)\,(\eta-1)\,(N-\eta+1)}{2\,(N-2)\,(N-1)\,(N+2)}$$
$$\Gamma_{\{3\}} = \frac{N\,(N-\eta)\,(\eta-1)\,(N-\eta+1)}{2\,(N-2)\,(N-1)\,(N+2)}$$
$$\Gamma_{\{4\}} = \frac{N\,(N-\eta)\,(\eta-1)\,(N-\eta+1)}{2\,(N-2)\,(N-1)\,(N+2)}$$
$$\Gamma_{\{1,2\}} = \frac{N\,(N-\eta)\,(N-\eta+1)}{(N-2)\,(N-1)\,(N+2)}$$
$$\Gamma_{\{1,3\}} = -\frac{N\,(N-\eta+1)\,(N\eta-3N+2\eta-2)}{4\,(N-2)\,(N-1)\,(N+2)}$$
$$\Gamma_{\{1,4\}} = -\frac{N\,(N-\eta+1)\,(N\eta-3N+2\eta-2)}{4\,(N-2)\,(N-1)\,(N+2)}$$
$$\Gamma_{\{2,3\}} = -\frac{N\,(N-\eta+1)\,(N\eta-3N+2\eta-2)}{4\,(N-2)\,(N-1)\,(N+2)}$$
$$\Gamma_{\{2,4\}} = -\frac{N\,(N-\eta+1)\,(N\eta-3N+2\eta-2)}{4\,(N-2)\,(N-1)\,(N+2)}$$
$$\Gamma_{\{3,4\}} = \frac{N\,(N-\eta)\,(N-\eta+1)}{(N-2)\,(N-1)\,(N+2)}$$
$$\Gamma_{\{1,2,3\}} = -\frac{N\,(N+1)\,(N-\eta+1)}{(N-2)\,(N-1)\,(N+2)}$$
$$\Gamma_{\{1,2,4\}} = -\frac{N\,(N+1)\,(N-\eta+1)}{(N-2)\,(N-1)\,(N+2)}$$
$$\Gamma_{\{1,3,4\}} = -\frac{N\,(N+1)\,(N-\eta+1)}{(N-2)\,(N-1)\,(N+2)}$$
$$\Gamma_{\{2,3,4\}} = -\frac{N\,(N+1)\,(N-\eta+1)}{(N-2)\,(N-1)\,(N+2)}$$
$$\Gamma_{\{1,2,3,4\}} = \frac{N\,(N+1)}{(N-2)\,(N-1)}$$

$k = 2$**, class V,** $(12, 12; 12, 13)$ $\quad L_+ = \{3\}$, $L_- = \{2\}$, $L_0 = \{1\}$.

$$A_{\emptyset} = \frac{(\eta-1)\,(N-\eta+1)\,(2N-\eta+1)}{4N\,(N+2)}$$
$$A_{\{1\}} = \frac{(N-\eta+1)\left(N\eta+5N-\eta^2+1\right)}{4N\,(N+2)}$$

$$\Gamma_{\emptyset} = \frac{(\eta-1)(N-\eta+1)(2N-\eta+1)}{4N(N+2)}$$
$$\Gamma_{\{1\}} = -\frac{(N-\eta+1)(N\eta-7N+2\eta-2)}{4N(N+2)}$$

$k=2$**, class V',** $(12,12;13,23)$ $\quad L_+=\{2\}$, $L_-=\{1\}$, $L_0=\{3\}$.

$$A_{\emptyset} = -\frac{(\eta-1)(\eta+1)(N-\eta+1)}{4(N-1)(N+2)}$$
$$A_{\{3\}} = -\frac{(N-\eta+1)\left(N\eta-3N+\eta^2+3\right)}{4(N-1)(N+2)}$$

$$\Gamma_{\emptyset} = -\frac{(\eta-1)(\eta+1)(N-\eta+1)}{4(N-1)(N+2)}$$
$$\Gamma_{\{3\}} = -\frac{(N-\eta+1)(N\eta-3N+4)}{4(N-1)(N+2)}$$

$k=2$**, class V",** $(12,13;12,23)$ $\quad L_+=\{2\}$, $L_-=\{1\}$, $L_0=\{3\}$.

$$A_{\emptyset} = \frac{(\eta-1)(N-\eta+1)\left(2N^2+\eta-1\right)}{4N(N-1)(N+2)}$$
$$A_{\{3\}} = \frac{(\eta+1)(N-\eta+1)\left(2N^2-N+\eta-1\right)}{4N(N-1)(N+2)}$$

$$\Gamma_{\emptyset} = \frac{(\eta-1)(N-\eta+1)\left(2N^2+\eta-1\right)}{4N(N-1)(N+2)}$$
$$\Gamma_{\{3\}} = \frac{(N-\eta+1)\left(4N^2-N\eta-N+2\eta-2\right)}{4N(N-1)(N+2)}$$

$k=2$**, class VI,** $(12,12;13,14)$ $\quad L_+=\{4\}$, $L_-=\{3\}$, $L_0=\{1,2\}$.

$$A_{\emptyset} = \frac{(\eta-1)(N-\eta+1)(N\eta-3N+2\eta-2)}{4N(N-1)(N+2)}$$
$$A_{\{1\}} = \frac{(N-\eta+1)\left(N\eta^2-N\eta-6N+2\eta^2-2\eta\right)}{4N(N-1)(N+2)}$$
$$A_{\{2\}} = \frac{(N+\eta)(N-\eta+1)(N\eta-3N+2\eta-2)}{4N(N-1)(N+2)}$$
$$A_{\{1,2\}} = \frac{(N-\eta+1)\left(N^2\eta+5N^2+N\eta^2+2N\eta-3N+2\eta^2-2\right)}{4N(N-1)(N+2)}$$

$$\Gamma_{\emptyset} = \frac{(\eta-1)(N-\eta+1)(N\eta-3N+2\eta-2)}{4N(N-1)(N+2)}$$
$$\Gamma_{\{1\}} = \frac{(N-\eta+1)(3N\eta-9N+2\eta-2)}{4N(N-1)(N+2)}$$

$$\Gamma_{\{2\}} = \frac{(N+1)(N-\eta+1)(N\eta-3N+2\eta-2)}{4N(N-1)(N+2)}$$
$$\Gamma_{\{1,2\}} = \frac{2(N+1)(N-\eta+1)}{(N-1)(N+2)}$$

$k=2$, class VI', $(12,12;13,34)$ $\quad L_+=\{4\}$, $L_-=\{1\}$, $L_0=\{2,3\}$.

$$A_{\emptyset} = \frac{(\eta-1)(N-\eta+1)(2N-\eta+1)}{4N(N-1)(N+2)}$$
$$A_{\{2\}} = \frac{(\eta+1)(N-\eta+1)^2}{4N(N-1)(N+2)}$$
$$A_{\{3\}} = \frac{(N-\eta+1)\left(4N-\eta^2+2\eta-1\right)}{4N(N-1)(N+2)}$$
$$A_{\{2,3\}} = -\frac{(N-\eta+1)\left(4N^2+N\eta-3N+\eta^2-1\right)}{4N(N-1)(N+2)}$$

$$\Gamma_{\emptyset} = \frac{(\eta-1)(N-\eta+1)(2N-\eta+1)}{4N(N-1)(N+2)}$$
$$\Gamma_{\{2\}} = -\frac{(N-\eta+1)(N\eta-3N+2\eta-2)}{4N(N-1)(N+2)}$$
$$\Gamma_{\{3\}} = -\frac{(\eta-3)(N-\eta+1)}{2(N-1)(N+2)}$$
$$\Gamma_{\{2,3\}} = -\frac{(N+1)(N-\eta+1)}{(N-1)(N+2)}$$

$k=2$, class VI", $(12,13;12,14)$ $\quad L_+=\{4\}$, $L_-=\{3\}$, $L_0=\{1,2\}$.

$$A_{\emptyset} = \frac{(\eta-1)(N-\eta+1)\left(2N^2-2N\eta+2N-\eta+1\right)}{4N(N-1)(N+2)}$$
$$A_{\{1\}} = \frac{(N-\eta+1)\left(N^2\eta+5N^2-2N\eta^2+2N-\eta^2+2\eta-1\right)}{4N(N-1)(N+2)}$$
$$A_{\{2\}} = \frac{(\eta+1)(N-\eta+1)\left(N^2-2N\eta+3N-\eta+1\right)}{4N(N-1)(N+2)}$$
$$A_{\{1,2\}} = -\frac{(\eta+1)(N-\eta+1)(2N\eta+N+\eta-1)}{4N(N-1)(N+2)}$$

$$\Gamma_{\emptyset} = \frac{(\eta-1)(N-\eta+1)\left(2N^2-2N\eta+2N-\eta+1\right)}{4N(N-1)(N+2)}$$
$$\Gamma_{\{1\}} = -\frac{(N-\eta+1)(N\eta-7N+4\eta-4)}{4(N-1)(N+2)}$$
$$\Gamma_{\{2\}} = -\frac{(N+1)(N-\eta+1)(N\eta-3N+2\eta-2)}{4N(N-1)(N+2)}$$
$$\Gamma_{\{1,2\}} = -\frac{2(N+1)(N-\eta+1)}{(N-1)(N+2)}$$

$k=2$**, class VI"',** $(12,13;12,34)\quad L_+=\{4\},\ L_-=\{1\},\ L_0=\{2,3\}.$

$$
\begin{aligned}
A_{\emptyset} &= -\frac{(\eta-1)(N-\eta+1)(2N-\eta+1)}{4(N-1)(N+2)}\\
A_{\{2\}} &= -\frac{(\eta+1)(N-\eta+1)^2}{4(N-1)(N+2)}\\
A_{\{3\}} &= -\frac{(\eta+1)(N-\eta+1)(2N-\eta+1)}{4(N-1)(N+2)}\\
A_{\{2,3\}} &= -\frac{(\eta+1)(N-\eta-1)(N-\eta+1)}{4(N-1)(N+2)}
\end{aligned}
$$

$$
\begin{aligned}
\Gamma_{\emptyset} &= -\frac{(\eta-1)(N-\eta+1)(2N-\eta+1)}{4(N-1)(N+2)}\\
\Gamma_{\{2\}} &= \frac{(N-\eta+1)(N\eta-3N+2\eta-2)}{4(N-1)(N+2)}\\
\Gamma_{\{3\}} &= -\frac{(N-\eta+1)(2N-\eta+1)}{2(N-1)(N+2)}\\
\Gamma_{\{2,3\}} &= \frac{(N+1)(N-\eta+1)}{(N-1)(N+2)}
\end{aligned}
$$

$k=2$**, class VI"",** $(12,13;14,34)\quad L_+=\{2\},\ L_-=\{1\},\ L_0=\{3,4\}.$

$$
\begin{aligned}
A_{\emptyset} &= -\frac{(N+1)(\eta-1)^2(N-\eta+1)}{4N(N-1)(N+2)}\\
A_{\{3\}} &= -\frac{(\eta+1)(N-\eta+1)(N\eta-2N+\eta-1)}{4N(N-1)(N+2)}\\
A_{\{4\}} &= -\frac{(N-\eta+1)\left(N^2\eta-3N^2+N\eta^2-N+\eta^2-2\eta+1\right)}{4N(N-1)(N+2)}\\
A_{\{3,4\}} &= -\frac{(N+1)(\eta+1)(N-\eta+1)(N+\eta-1)}{4N(N-1)(N+2)}
\end{aligned}
$$

$$
\begin{aligned}
\Gamma_{\emptyset} &= -\frac{(N+1)(\eta-1)^2(N-\eta+1)}{4N(N-1)(N+2)}\\
\Gamma_{\{3\}} &= -\frac{(N-\eta+1)(N\eta-3N+2\eta-2)}{4N(N-1)(N+2)}\\
\Gamma_{\{4\}} &= -\frac{(N-\eta+1)(N\eta-3N+2\eta-2)}{4(N-1)(N+2)}\\
\Gamma_{\{3,4\}} &= -\frac{(N+1)(N-\eta+1)}{(N-1)(N+2)}
\end{aligned}
$$

$k=2$**, class VII,** $(12,13;24,45)\quad L_+=\{5\},\ L_-=\{3\},\ L_0=\{1,2,4\}.$

$$
A_{\emptyset} = \frac{(\eta-1)(N-\eta+1)\left(2N^2-3N\eta+3N-2\eta+2\right)}{4N(N-2)(N-1)(N+2)}
$$

$$
\begin{aligned}
A_{\{1\}} &= \frac{(N-\eta+1)\left(4N^2-3N\eta^2+2N\eta+N-2\eta^2+4\eta-2\right)}{4N(N-2)(N-1)(N+2)} \\
A_{\{2\}} &= \frac{(\eta+1)(N-\eta+1)\left(N^2-3N\eta+5N-2\eta+2\right)}{4N(N-2)(N-1)(N+2)} \\
A_{\{4\}} &= \frac{(N-\eta+1)\left(4N^2-3N\eta^2+2N\eta+N-2\eta^2+4\eta-2\right)}{4N(N-2)(N-1)(N+2)} \\
A_{\{1,2\}} &= -\frac{(\eta+1)(N-\eta+1)\left(N^2+3N\eta-N+2\eta-2\right)}{4N(N-2)(N-1)(N+2)} \\
A_{\{1,4\}} &= -\frac{(N-\eta+1)\left(4N^3+2N^2\eta-6N^2+3N\eta^2+2N\eta-5N+2\eta^2-4\eta+2\right)}{4N(N-2)(N-1)(N+2)} \\
A_{\{2,4\}} &= -\frac{(\eta+1)(N-\eta+1)\left(N^2+3N\eta-N+2\eta-2\right)}{4N(N-2)(N-1)(N+2)} \\
A_{\{1,2,4\}} &= \frac{(\eta+1)\left(N^3-2N^2+3N\eta^2-2N\eta-N+2\eta^2-4\eta+2\right)}{4N(N-2)(N-1)(N+2)}
\end{aligned}
$$

$$
\begin{aligned}
\Gamma_{\emptyset} &= \frac{(\eta-1)(N-\eta+1)\left(2N^2-3N\eta+3N-2\eta+2\right)}{4N(N-2)(N-1)(N+2)} \\
\Gamma_{\{1\}} &= -\frac{(N-\eta+1)(N\eta-3N+2\eta-2)}{2(N-2)(N-1)(N+2)} \\
\Gamma_{\{2\}} &= -\frac{(N-\eta+1)(N\eta-3N+2\eta-2)}{4N(N-2)(N-1)} \\
\Gamma_{\{4\}} &= -\frac{(N-\eta+1)(N\eta-3N+2\eta-2)}{2(N-2)(N-1)(N+2)} \\
\Gamma_{\{1,2\}} &= -\frac{2(N+1)(N-\eta+1)}{(N-2)(N-1)(N+2)} \\
\Gamma_{\{1,4\}} &= -\frac{N(N+1)(N-\eta+1)}{(N-2)(N-1)(N+2)} \\
\Gamma_{\{2,4\}} &= -\frac{2(N+1)(N-\eta+1)}{(N-2)(N-1)(N+2)} \\
\Gamma_{\{1,2,4\}} &= \frac{N(N+1)}{(N-2)(N-1)}
\end{aligned}
$$

$k=2$**, class VII',** $(12,12;34,35)\quad L_+=\{5\},\ L_-=\{4\},\ L_0=\{1,2,3\}.$

$$
\begin{aligned}
A_{\emptyset} &= -\frac{(\eta-1)(N-\eta+1)(N\eta-3N+2\eta-2)}{2N(N-2)(N-1)(N+2)} \\
A_{\{1\}} &= -\frac{(N+2\eta)(N-\eta+1)(N\eta-3N+2\eta-2)}{4N(N-2)(N-1)(N+2)} \\
A_{\{2\}} &= -\frac{(N+2\eta)(N-\eta+1)(N\eta-3N+2\eta-2)}{4N(N-2)(N-1)(N+2)} \\
A_{\{3\}} &= -\frac{(N-\eta+1)\left(N\eta^2-5N+2\eta^2-4\eta+2\right)}{2N(N-2)(N-1)(N+2)} \\
A_{\{1,2\}} &= -\frac{(\eta+1)(N-\eta+1)\left(N^2+N\eta-N+2\eta-2\right)}{2N(N-2)(N-1)(N+2)}
\end{aligned}
$$

$$A_{\{1,3\}} = -\frac{(N-\eta+1)\left(N^2\eta+5N^2+2N\eta^2+4N\eta-10N+4\eta^2-4\eta\right)}{4N(N-2)(N-1)(N+2)}$$
$$A_{\{2,3\}} = -\frac{(N-\eta+1)\left(N^2\eta+5N^2+2N\eta^2+4N\eta-10N+4\eta^2-4\eta\right)}{4N(N-2)(N-1)(N+2)}$$
$$A_{\{1,2,3\}} = \frac{2N^4-N^3\eta-3N^3+4N^2\eta-4N^2+N\eta^3+N\eta^2-5N\eta+3N+2\eta^3-2\eta^2-2\eta+2}{2N(N-2)(N-1)(N+2)}$$

$$\Gamma_{\emptyset} = -\frac{(\eta-1)(N-\eta+1)(N\eta-3N+2\eta-2)}{2N(N-2)(N-1)(N+2)}$$
$$\Gamma_{\{1\}} = -\frac{(N-\eta+1)(N\eta-3N+2\eta-2)}{4N(N-2)(N-1)}$$
$$\Gamma_{\{2\}} = -\frac{(N-\eta+1)(N\eta-3N+2\eta-2)}{4N(N-2)(N-1)}$$
$$\Gamma_{\{3\}} = -\frac{2(\eta-2)(N-\eta+1)}{(N-2)(N-1)(N+2)}$$
$$\Gamma_{\{1,2\}} = -\frac{2(N+1)(N-\eta+1)}{(N-2)(N-1)(N+2)}$$
$$\Gamma_{\{1,3\}} = -\frac{2(N+1)(N-\eta+1)}{(N-2)(N-1)(N+2)}$$
$$\Gamma_{\{2,3\}} = -\frac{2(N+1)(N-\eta+1)}{(N-2)(N-1)(N+2)}$$
$$\Gamma_{\{1,2,3\}} = \frac{N(N+1)}{(N-2)(N-1)}$$

$k=2$**, class VII",** $(12,13;24,35)$ $\quad L_+=\{5\}$, $L_-=\{4\}$, $L_0=\{1,2,3\}$.

$$A_{\emptyset} = \frac{(\eta-1)(N-\eta+1)(N\eta-3N+2\eta-2)}{4(N-2)(N-1)(N+2)}$$
$$A_{\{1\}} = \frac{(N-\eta+1)(N+\eta-1)(N\eta-3N+2\eta-2)}{4(N-2)(N-1)(N+2)}$$
$$A_{\{2\}} = \frac{(\eta+1)(N-\eta+1)(N\eta-3N+2\eta-2)}{4(N-2)(N-1)(N+2)}$$
$$A_{\{3\}} = \frac{(\eta+1)(N-\eta+1)(N\eta-3N+2\eta-2)}{4(N-2)(N-1)(N+2)}$$
$$A_{\{1,2\}} = \frac{(\eta+1)(N-\eta+1)\left(N^2+N\eta-N+2\eta-2\right)}{4(N-2)(N-1)(N+2)}$$
$$A_{\{1,3\}} = \frac{(\eta+1)(N-\eta+1)\left(N^2+N\eta-N+2\eta-2\right)}{4(N-2)(N-1)(N+2)}$$
$$A_{\{2,3\}} = \frac{(\eta+1)(N-\eta+1)(N\eta-N+2\eta+2)}{4(N-2)(N-1)(N+2)}$$
$$A_{\{1,2,3\}} = \frac{(\eta+1)(N-\eta-1)\left(N^2+N\eta-N+2\eta-2\right)}{4(N-2)(N-1)(N+2)}$$

$$\Gamma_{\emptyset} = \frac{(\eta-1)(N-\eta+1)(N\eta-3N+2\eta-2)}{4(N-2)(N-1)(N+2)}$$

$$\Gamma_{\{1\}} = \frac{N\left(N-\eta+1\right)\left(N\eta-3N+2\eta-2\right)}{4\left(N-2\right)\left(N-1\right)\left(N+2\right)}$$
$$\Gamma_{\{2\}} = \frac{\left(N-\eta+1\right)\left(N\eta-3N+2\eta-2\right)}{2\left(N-2\right)\left(N-1\right)\left(N+2\right)}$$
$$\Gamma_{\{3\}} = \frac{\left(N-\eta+1\right)\left(N\eta-3N+2\eta-2\right)}{2\left(N-2\right)\left(N-1\right)\left(N+2\right)}$$
$$\Gamma_{\{1,2\}} = \frac{N\left(N+1\right)\left(N-\eta+1\right)}{\left(N-2\right)\left(N-1\right)\left(N+2\right)}$$
$$\Gamma_{\{1,3\}} = \frac{N\left(N+1\right)\left(N-\eta+1\right)}{\left(N-2\right)\left(N-1\right)\left(N+2\right)}$$
$$\Gamma_{\{2,3\}} = \frac{2\left(N+1\right)\left(N-\eta+1\right)}{\left(N-2\right)\left(N-1\right)\left(N+2\right)}$$
$$\Gamma_{\{1,2,3\}} = -\frac{N\left(N+1\right)}{\left(N-2\right)\left(N-1\right)}$$

$k=2$**, class VII"',** $(12,34;12,35)$ $\quad L_+=\{5\}$, $L_-=\{4\}$, $L_0=\{1,2,3\}$.

$$A_{\emptyset} = \frac{N\left(N-\eta\right)\left(\eta-1\right)\left(N-\eta+1\right)}{2\left(N-2\right)\left(N-1\right)\left(N+2\right)}$$
$$A_{\{1\}} = \frac{N\left(\eta+1\right)\left(N-2\eta+2\right)\left(N-\eta+1\right)}{4\left(N-2\right)\left(N-1\right)\left(N+2\right)}$$
$$A_{\{2\}} = \frac{N\left(\eta+1\right)\left(N-2\eta+2\right)\left(N-\eta+1\right)}{4\left(N-2\right)\left(N-1\right)\left(N+2\right)}$$
$$A_{\{3\}} = \frac{N\left(N-\eta\right)\left(\eta+1\right)\left(N-\eta+1\right)}{2\left(N-2\right)\left(N-1\right)\left(N+2\right)}$$
$$A_{\{1,2\}} = -\frac{N\eta\left(\eta+1\right)\left(N-\eta+1\right)}{2\left(N-2\right)\left(N-1\right)\left(N+2\right)}$$
$$A_{\{1,3\}} = \frac{N\left(\eta+1\right)\left(N-2\eta-2\right)\left(N-\eta+1\right)}{4\left(N-2\right)\left(N-1\right)\left(N+2\right)}$$
$$A_{\{2,3\}} = \frac{N\left(\eta+1\right)\left(N-2\eta-2\right)\left(N-\eta+1\right)}{4\left(N-2\right)\left(N-1\right)\left(N+2\right)}$$
$$A_{\{1,2,3\}} = -\frac{N\eta\left(\eta+1\right)\left(N-\eta-1\right)}{2\left(N-2\right)\left(N-1\right)\left(N+2\right)}$$

$$\Gamma_{\emptyset} = \frac{N\left(N-\eta\right)\left(\eta-1\right)\left(N-\eta+1\right)}{2\left(N-2\right)\left(N-1\right)\left(N+2\right)}$$
$$\Gamma_{\{1\}} = -\frac{N\left(N-\eta+1\right)\left(N\eta-3N+2\eta-2\right)}{4\left(N-2\right)\left(N-1\right)\left(N+2\right)}$$
$$\Gamma_{\{2\}} = -\frac{N\left(N-\eta+1\right)\left(N\eta-3N+2\eta-2\right)}{4\left(N-2\right)\left(N-1\right)\left(N+2\right)}$$
$$\Gamma_{\{3\}} = \frac{N\left(N-\eta\right)\left(N-\eta+1\right)}{\left(N-2\right)\left(N-1\right)\left(N+2\right)}$$
$$\Gamma_{\{1,2\}} = -\frac{N\left(N+1\right)\left(N-\eta+1\right)}{\left(N-2\right)\left(N-1\right)\left(N+2\right)}$$

$$\Gamma_{\{1,3\}} = -\frac{N\,(N+1)\,(N-\eta+1)}{(N-2)\,(N-1)\,(N+2)}$$
$$\Gamma_{\{2,3\}} = -\frac{N\,(N+1)\,(N-\eta+1)}{(N-2)\,(N-1)\,(N+2)}$$
$$\Gamma_{\{1,2,3\}} = \frac{N\,(N+1)}{(N-2)\,(N-1)}$$

$k=2$**, class VIII,** $(12,34;13,13)\quad L_+=\{1,2\},\ L_-=\{3,4\},\ L_0=\emptyset.$

$$A_\emptyset = -\frac{(N-\eta+1)\,(N\eta-3N+4)}{4\,(N-1)\,(N+2)}$$

$$\Gamma_\emptyset = -\frac{(N-\eta+1)\,(N\eta-3N+4)}{4\,(N-1)\,(N+2)}$$

$k=2$**, class VIII',** $(12,12;12,34)\quad L_+=\{3,4\},\ L_-=\{1,2\},\ L_0=\emptyset.$

$$A_\emptyset = \frac{N-\eta+1}{N+2}$$

$$\Gamma_\emptyset = \frac{N-\eta+1}{N+2}$$

$k=2$**, class VIII",** $(12,13;12,24)\quad L_+=\{2,4\},\ L_-=\{1,3\},\ L_0=\emptyset.$

$$A_\emptyset = \frac{(N-\eta+1)\left(4N^2-N\eta-N+2\eta-2\right)}{4N\,(N-1)\,(N+2)}$$

$$\Gamma_\emptyset = \frac{(N-\eta+1)\left(4N^2-N\eta-N+2\eta-2\right)}{4N\,(N-1)\,(N+2)}$$

$k=2$**, class VIII"',** $(12,13;13,34)\quad L_+=\{2,4\},\ L_-=\{1,3\},\ L_0=\emptyset.$

$$A_\emptyset = -\frac{(N-\eta+1)\,(N\eta-3N+2\eta-2)}{4N\,(N+2)}$$

$$\Gamma_\emptyset = -\frac{(N-\eta+1)\,(N\eta-3N+2\eta-2)}{4N\,(N+2)}$$

$k=2$**, class IX,** $(12,13;14,15)\quad L_+=\{2,5\},\ L_-=\{3,4\},\ L_0=\{1\}.$

$$A_\emptyset=\frac{(N+1)\,(N-\eta+1)\,(N\eta-3N+2\eta-2)}{4N\,(N-1)\,(N+2)}$$
$$A_{\{1\}}=\frac{(N+1)\,(N-\eta+1)\,(N\eta+5N+2\eta-2)}{4N\,(N-1)\,(N+2)}$$

$$\Gamma_\emptyset=\frac{(N+1)\,(N-\eta+1)\,(N\eta-3N+2\eta-2)}{4N\,(N-1)\,(N+2)}$$
$$\Gamma_{\{1\}}=\frac{2\,(N+1)\,(N-\eta+1)}{(N-1)\,(N+2)}$$

$k=2$**, class IX',** $(12,12;13,45)\quad L_+=\{4,5\},\ L_-=\{1,3\},\ L_0=\{2\}.$

$$A_\emptyset=\frac{(\eta-3)\,(N-\eta+1)}{2\,(N-1)\,(N+2)}$$
$$A_{\{2\}}=\frac{(N-\eta+1)\,(2N+\eta-1)}{2\,(N-1)\,(N+2)}$$

$$\Gamma_\emptyset=\frac{(\eta-3)\,(N-\eta+1)}{2\,(N-1)\,(N+2)}$$
$$\Gamma_{\{2\}}=\frac{(N+1)\,(N-\eta+1)}{(N-1)\,(N+2)}$$

$k=2$**, class IX",** $(12,13;12,45)\quad L_+=\{4,5\},\ L_-=\{1,3\},\ L_0=\{2\}.$

$$A_\emptyset=\frac{(N-\eta+1)\,(2N-\eta+1)}{2\,(N-1)\,(N+2)}$$
$$A_{\{2\}}=-\frac{(\eta+1)\,(N-\eta+1)}{2\,(N-1)\,(N+2)}$$

$$\Gamma_\emptyset=\frac{(N-\eta+1)\,(2N-\eta+1)}{2\,(N-1)\,(N+2)}$$
$$\Gamma_{\{2\}}=-\frac{(N+1)\,(N-\eta+1)}{(N-1)\,(N+2)}$$

$k=2$**, class IX"',** $(12,13;14,35)\quad L_+=\{2,5\},\ L_-=\{1,4\},\ L_0=\{3\}.$

$$A_\emptyset=-\frac{(N-\eta+1)\,(N\eta-3N+2\eta-2)}{4\,(N-1)\,(N+2)}$$
$$A_{\{3\}}=-\frac{(\eta+1)\,(N-\eta+1)}{4\,(N-1)}$$

$$\Gamma_\emptyset=-\frac{(N-\eta+1)\,(N\eta-3N+2\eta-2)}{4\,(N-1)\,(N+2)}$$
$$\Gamma_{\{3\}}=-\frac{(N+1)\,(N-\eta+1)}{(N-1)\,(N+2)}$$

$k=2$**, class IX"",** $(12,13;14,45)$ $\quad L_+=\{2,5\},\ L_-=\{1,3\},\ L_0=\{4\}.$

$$A_\emptyset = \frac{(N-\eta+1)\,(N\eta-3N+2\eta-2)}{4N\,(N-1)\,(N+2)}$$
$$A_{\{4\}} = \frac{(N-\eta+1)\left(4N^2+N\eta+N+2\eta-2\right)}{4N\,(N-1)\,(N+2)}$$

$$\Gamma_\emptyset = \frac{(N-\eta+1)\,(N\eta-3N+2\eta-2)}{4N\,(N-1)\,(N+2)}$$
$$\Gamma_{\{4\}} = \frac{(N+1)\,(N-\eta+1)}{(N-1)\,(N+2)}$$

$k=2$**, class X,** $(12,13;45,46)$ $\quad L_+=\{2,6\},\ L_-=\{3,5\},\ L_0=\{1,4\}.$

$$A_\emptyset = -\frac{(N-\eta+1)\,(N\eta-3N+2\eta-2)}{4N\,(N-2)\,(N-1)}$$
$$A_{\{1\}} = -\frac{(N-\eta+1)\left(N^2\eta+5N^2+4N\eta+4\eta-4\right)}{4N\,(N-2)\,(N-1)\,(N+2)}$$
$$A_{\{4\}} = -\frac{(N-\eta+1)\left(N^2\eta+5N^2+4N\eta+4\eta-4\right)}{4N\,(N-2)\,(N-1)\,(N+2)}$$
$$A_{\{1,4\}} = \frac{4N^4-N^3\eta-N^3+N^2\eta^2+8N^2\eta-13N^2+4N\eta^2-4N+4\eta^2-8\eta+4}{4N\,(N-2)\,(N-1)\,(N+2)}$$

$$\Gamma_\emptyset = -\frac{(N-\eta+1)\,(N\eta-3N+2\eta-2)}{4N\,(N-2)\,(N-1)}$$
$$\Gamma_{\{1\}} = -\frac{2\,(N+1)\,(N-\eta+1)}{(N-2)\,(N-1)\,(N+2)}$$
$$\Gamma_{\{4\}} = -\frac{2\,(N+1)\,(N-\eta+1)}{(N-2)\,(N-1)\,(N+2)}$$
$$\Gamma_{\{1,4\}} = \frac{N\,(N+1)}{(N-2)\,(N-1)}$$

$k=2$**, class X',** $(12,12;34,56)$ $\quad L_+=\{5,6\},\ L_-=\{3,4\},\ L_0=\{1,2\}.$

$$A_\emptyset = -\frac{2\,(\eta-2)\,(N-\eta+1)}{(N-2)\,(N-1)\,(N+2)}$$
$$A_{\{1\}} = -\frac{2\,(N-\eta+1)\,(N+\eta-1)}{(N-2)\,(N-1)\,(N+2)}$$
$$A_{\{2\}} = -\frac{2\,(N-\eta+1)\,(N+\eta-1)}{(N-2)\,(N-1)\,(N+2)}$$
$$A_{\{1,2\}} = \frac{N^3-N^2+2N\eta-2N+2\eta^2-2\eta}{(N-2)\,(N-1)\,(N+2)}$$

$$\Gamma_{\emptyset} = -\frac{2\left(\eta-2\right)\left(N-\eta+1\right)}{\left(N-2\right)\left(N-1\right)\left(N+2\right)}$$
$$\Gamma_{\{1\}} = -\frac{2\left(N+1\right)\left(N-\eta+1\right)}{\left(N-2\right)\left(N-1\right)\left(N+2\right)}$$
$$\Gamma_{\{2\}} = -\frac{2\left(N+1\right)\left(N-\eta+1\right)}{\left(N-2\right)\left(N-1\right)\left(N+2\right)}$$
$$\Gamma_{\{1,2\}} = \frac{N\left(N+1\right)}{\left(N-2\right)\left(N-1\right)}$$

$k=2$**, class X",** $(12,13;24,56)$ $\quad L_+=\{5,6\}$, $L_-=\{3,4\}$, $L_0=\{1,2\}$.

$$A_{\emptyset} = \frac{\left(N-\eta+1\right)\left(N\eta-3N+2\eta-2\right)}{2\left(N-2\right)\left(N-1\right)\left(N+2\right)}$$
$$A_{\{1\}} = \frac{\left(N-\eta+1\right)\left(2N^2+N\eta-N+2\eta-2\right)}{2\left(N-2\right)\left(N-1\right)\left(N+2\right)}$$
$$A_{\{2\}} = \frac{\left(\eta+1\right)\left(N-\eta+1\right)}{2\left(N-2\right)\left(N-1\right)}$$
$$A_{\{1,2\}} = -\frac{\left(\eta+1\right)\left(N^2+N\eta-N+2\eta-2\right)}{2\left(N-2\right)\left(N-1\right)\left(N+2\right)}$$

$$\Gamma_{\emptyset} = \frac{\left(N-\eta+1\right)\left(N\eta-3N+2\eta-2\right)}{2\left(N-2\right)\left(N-1\right)\left(N+2\right)}$$
$$\Gamma_{\{1\}} = \frac{N\left(N+1\right)\left(N-\eta+1\right)}{\left(N-2\right)\left(N-1\right)\left(N+2\right)}$$
$$\Gamma_{\{2\}} = \frac{2\left(N+1\right)\left(N-\eta+1\right)}{\left(N-2\right)\left(N-1\right)\left(N+2\right)}$$
$$\Gamma_{\{1,2\}} = -\frac{N\left(N+1\right)}{\left(N-2\right)\left(N-1\right)}$$

$k=2$**, class X"',** $(12,34;12,56)$ $\quad L_+=\{5,6\}$, $L_-=\{3,4\}$, $L_0=\{1,2\}$.

$$A_{\emptyset} = \frac{N\left(N-\eta\right)\left(N-\eta+1\right)}{\left(N-2\right)\left(N-1\right)\left(N+2\right)}$$
$$A_{\{1\}} = -\frac{N\left(\eta+1\right)\left(N-\eta+1\right)}{\left(N-2\right)\left(N-1\right)\left(N+2\right)}$$
$$A_{\{2\}} = -\frac{N\left(\eta+1\right)\left(N-\eta+1\right)}{\left(N-2\right)\left(N-1\right)\left(N+2\right)}$$
$$A_{\{1,2\}} = \frac{N\eta\left(\eta+1\right)}{\left(N-2\right)\left(N-1\right)\left(N+2\right)}$$

$$\Gamma_{\emptyset} = \frac{N\left(N-\eta\right)\left(N-\eta+1\right)}{\left(N-2\right)\left(N-1\right)\left(N+2\right)}$$
$$\Gamma_{\{1\}} = -\frac{N\left(N+1\right)\left(N-\eta+1\right)}{\left(N-2\right)\left(N-1\right)\left(N+2\right)}$$

$$\Gamma_{\{2\}} = -\frac{N\,(N+1)\,(N-\eta+1)}{(N-2)\,(N-1)\,(N+2)}$$
$$\Gamma_{\{1,2\}} = \frac{N\,(N+1)}{(N-2)\,(N-1)}$$

$k=2$**, class X"",** $(12,34;15,36)\quad L_+=\{2,6\},\ L_-=\{4,5\},\ L_0=\{1,3\}.$

$$A_{\emptyset} = \frac{N\,(N-\eta+1)\,(N\eta-3N+2\eta-2)}{4\,(N-2)\,(N-1)\,(N+2)}$$
$$A_{\{1\}} = \frac{N\,(\eta+1)\,(N-\eta+1)}{4\,(N-2)\,(N-1)}$$
$$A_{\{3\}} = \frac{N\,(\eta+1)\,(N-\eta+1)}{4\,(N-2)\,(N-1)}$$
$$A_{\{1,3\}} = \frac{N\,(\eta+1)\left(N^2-N\eta-N-2\eta-2\right)}{4\,(N-2)\,(N-1)\,(N+2)}$$

$$\Gamma_{\emptyset} = \frac{N\,(N-\eta+1)\,(N\eta-3N+2\eta-2)}{4\,(N-2)\,(N-1)\,(N+2)}$$
$$\Gamma_{\{1\}} = \frac{N\,(N+1)\,(N-\eta+1)}{(N-2)\,(N-1)\,(N+2)}$$
$$\Gamma_{\{3\}} = \frac{N\,(N+1)\,(N-\eta+1)}{(N-2)\,(N-1)\,(N+2)}$$
$$\Gamma_{\{1,3\}} = -\frac{N\,(N+1)}{(N-2)\,(N-1)}$$

$k=2$**, class XI,** $(12,45;16,13)\quad L_+=\{1,2,3\},\ L_-=\{4,5,6\},\ L_0=\emptyset.$

$$A_{\emptyset} = \frac{(N+1)\,(N-\eta+1)}{(N-1)\,(N+2)}$$

$$\Gamma_{\emptyset} = \frac{(N+1)\,(N-\eta+1)}{(N-1)\,(N+2)}$$

$k=2$**, class XII,** $(12,45;67,37)\quad L_+=\{1,2,3\},\ L_-=\{4,5,6\},\ L_0=\{7\}.$

$$A_{\emptyset} = -\frac{2\,(N+1)\,(N-\eta+1)}{(N-2)\,(N-1)\,(N+2)}$$
$$A_{\{7\}} = \frac{(N+1)\left(N^2+2\eta-2\right)}{(N-2)\,(N-1)\,(N+2)}$$

$$\Gamma_{\emptyset} = -\frac{2\,(N+1)\,(N-\eta+1)}{(N-2)\,(N-1)\,(N+2)}$$
$$\Gamma_{\{7\}} = \frac{N\,(N+1)}{(N-2)\,(N-1)}$$

$k = 2$**, class XII',** $(12, 34; 15, 67)$ $\quad L_+ = \{2, 6, 7\}$, $L_- = \{3, 4, 5\}$, $L_0 = \{1\}$.

$$A_\emptyset = -\frac{N\,(N+1)\,(N-\eta+1)}{(N-2)\,(N-1)\,(N+2)}$$
$$A_{\{1\}} = \frac{N\,(N+1)\,(\eta+1)}{(N-2)\,(N-1)\,(N+2)}$$

$$\Gamma_\emptyset = -\frac{N\,(N+1)\,(N-\eta+1)}{(N-2)\,(N-1)\,(N+2)}$$
$$\Gamma_{\{1\}} = \frac{N\,(N+1)}{(N-2)\,(N-1)}$$

$k = 2$**, class XIII,** $(12, 34; 78, 56)$ $\quad L_+ = \{1, 2, 5, 6\}$, $L_- = \{3, 4, 7, 8\}$, $L_0 = \emptyset$.

$$A_\emptyset = \frac{N\,(N+1)}{(N-2)\,(N-1)}$$

$$\Gamma_\emptyset = \frac{N\,(N+1)}{(N-2)\,(N-1)}$$

mixed $(1, 2)$**, class I,** $(1; 2; 12, 13)$ $\quad$ Class I: impossible charge, so no coefficients are defined.

mixed $(1, 2)$**, class C1,** $(1; 1; 12, 12)$ $\quad L_+ = \emptyset$, $L_- = \emptyset$, $L_0 = \{1, 2\}$.

$$A_\emptyset = \frac{\eta\,(\eta-1)\,(N-\eta+1)}{2N\,(N+2)}$$
$$A_{\{1\}} = \frac{(\eta-1)\,(\eta+1)\,(N-\eta+1)}{2N\,(N+2)}$$
$$A_{\{2\}} = \frac{(\eta-1)\,(N-\eta+1)\,(N+\eta+1)}{2N\,(N+2)}$$
$$A_{\{1,2\}} = \frac{(N-\eta+1)\left(N\eta+3N+\eta^2+\eta\right)}{2N\,(N+2)}$$

$$\Gamma_\emptyset = \frac{\eta\,(\eta-1)\,(N-\eta+1)}{2N\,(N+2)}$$
$$\Gamma_{\{1\}} = \frac{(\eta-1)\,(N-\eta+1)}{2N\,(N+2)}$$
$$\Gamma_{\{2\}} = \frac{(N+1)\,(\eta-1)\,(N-\eta+1)}{2N\,(N+2)}$$
$$\Gamma_{\{1,2\}} = \frac{(2N+1)\,(N-\eta+1)}{N\,(N+2)}$$

mixed $(1,2)$**, class C2,** $(1;1;23,23)$ $\quad L_+ = \emptyset,\ L_- = \emptyset,\ L_0 = \{1,2,3\}.$

$$
\begin{aligned}
A_{\emptyset} &= -\frac{\eta\,(\eta-1)\,(N-\eta+1)}{N\,(N-1)\,(N+2)} \\
A_{\{1\}} &= -\frac{(\eta-1)\,(\eta+1)\,(N-\eta+1)}{N\,(N-1)\,(N+2)} \\
A_{\{2\}} &= -\frac{(\eta-1)\,(N-\eta+1)\,(N+2\eta+1)}{2N\,(N-1)\,(N+2)} \\
A_{\{3\}} &= -\frac{(\eta-1)\,(N-\eta+1)\,(N+2\eta+1)}{2N\,(N-1)\,(N+2)} \\
A_{\{1,2\}} &= -\frac{(\eta+1)\,(N-\eta+1)\,(N+2\eta-1)}{2N\,(N-1)\,(N+2)} \\
A_{\{1,3\}} &= -\frac{(\eta+1)\,(N-\eta+1)\,(N+2\eta-1)}{2N\,(N-1)\,(N+2)} \\
A_{\{2,3\}} &= -\frac{(\eta+1)\,(N-\eta+1)\,(N+\eta-1)}{N\,(N-1)\,(N+2)} \\
A_{\{1,2,3\}} &= \frac{N^3-N^2\eta+N\eta-N+\eta^3-\eta}{N\,(N-1)\,(N+2)}
\end{aligned}
$$

$$
\begin{aligned}
\Gamma_{\emptyset} &= -\frac{\eta\,(\eta-1)\,(N-\eta+1)}{N\,(N-1)\,(N+2)} \\
\Gamma_{\{1\}} &= -\frac{(\eta-1)\,(N-\eta+1)}{N\,(N-1)\,(N+2)} \\
\Gamma_{\{2\}} &= -\frac{(N+1)\,(\eta-1)\,(N-\eta+1)}{2N\,(N-1)\,(N+2)} \\
\Gamma_{\{3\}} &= -\frac{(N+1)\,(\eta-1)\,(N-\eta+1)}{2N\,(N-1)\,(N+2)} \\
\Gamma_{\{1,2\}} &= -\frac{(N+1)\,(N-\eta+1)}{N\,(N-1)\,(N+2)} \\
\Gamma_{\{1,3\}} &= -\frac{(N+1)\,(N-\eta+1)}{N\,(N-1)\,(N+2)} \\
\Gamma_{\{2,3\}} &= -\frac{2\,(N-\eta+1)}{(N-1)\,(N+2)} \\
\Gamma_{\{1,2,3\}} &= \frac{N+1}{N-1}
\end{aligned}
$$

mixed $(1,2)$**, class C3,** $(1;2;13,23)$ $\quad L_+ = \emptyset,\ L_- = \emptyset,\ L_0 = \{1,2,3\}.$

$$
\begin{aligned}
A_{\emptyset} &= \frac{\eta\,(N+1)\,(\eta-1)\,(N-\eta+1)}{2N\,(N-1)\,(N+2)} \\
A_{\{1\}} &= \frac{(N+1)\,(\eta-1)\,(\eta+1)\,(N-\eta+1)}{2N\,(N-1)\,(N+2)} \\
A_{\{2\}} &= \frac{(N+1)\,(\eta-1)\,(\eta+1)\,(N-\eta+1)}{2N\,(N-1)\,(N+2)}
\end{aligned}
$$

$$A_{\{3\}} = \frac{(N+1)(N+\eta)(\eta-1)(N-\eta+1)}{2N(N-1)(N+2)}$$
$$A_{\{1,2\}} = \frac{\eta(N+1)(\eta+1)(N-\eta+1)}{2N(N-1)(N+2)}$$
$$A_{\{1,3\}} = \frac{(N+1)(\eta+1)(N-\eta+1)(N+\eta-1)}{2N(N-1)(N+2)}$$
$$A_{\{2,3\}} = \frac{(N+1)(\eta+1)(N-\eta+1)(N+\eta-1)}{2N(N-1)(N+2)}$$
$$A_{\{1,2,3\}} = \frac{(N+1)(N-\eta)(\eta+1)(N+\eta-1)}{2N(N-1)(N+2)}$$

$$\Gamma_{\emptyset} = \frac{\eta(N+1)(\eta-1)(N-\eta+1)}{2N(N-1)(N+2)}$$
$$\Gamma_{\{1\}} = \frac{(N+1)(\eta-1)(N-\eta+1)}{2N(N-1)(N+2)}$$
$$\Gamma_{\{2\}} = \frac{(N+1)(\eta-1)(N-\eta+1)}{2N(N-1)(N+2)}$$
$$\Gamma_{\{3\}} = \frac{(N+1)(\eta-1)(N-\eta+1)}{2(N-1)(N+2)}$$
$$\Gamma_{\{1,2\}} = \frac{(N+1)(N-\eta+1)}{N(N-1)(N+2)}$$
$$\Gamma_{\{1,3\}} = \frac{(N+1)(N-\eta+1)}{(N-1)(N+2)}$$
$$\Gamma_{\{2,3\}} = \frac{(N+1)(N-\eta+1)}{(N-1)(N+2)}$$
$$\Gamma_{\{1,2,3\}} = -\frac{N+1}{N-1}$$

mixed $(1,2)$**, class C4,** $(1;2;12,12)$ $\quad L_+ = \{1\}$, $L_- = \{2\}$, $L_0 = \emptyset$.

$$A_{\emptyset} = -\frac{(\eta-1)(N-\eta+1)}{2(N+2)}$$

$$\Gamma_{\emptyset} = -\frac{(\eta-1)(N-\eta+1)}{2(N+2)}$$

mixed $(1,2)$**, class C5,** $(1;1;12,13)$ $\quad L_+ = \{3\}$, $L_- = \{2\}$, $L_0 = \{1\}$.

$$A_{\emptyset} = \frac{(N+1)(\eta-1)(N-\eta+1)}{2N(N+2)}$$
$$A_{\{1\}} = \frac{(N-\eta+1)(N\eta+3N+\eta+1)}{2N(N+2)}$$

$$\Gamma_{\emptyset} = \frac{(N+1)(\eta-1)(N-\eta+1)}{2N(N+2)}$$
$$\Gamma_{\{1\}} = \frac{(2N+1)(N-\eta+1)}{N(N+2)}$$

mixed $(1,2)$**, class C6,** $(1;1;12,23)$ $\quad L_+ = \{3\}$, $L_- = \{1\}$, $L_0 = \{2\}$.

$$A_\emptyset = 0$$
$$A_{\{2\}} = -\frac{N-\eta+1}{N+2}$$

$$\Gamma_\emptyset = 0$$
$$\Gamma_{\{2\}} = -\frac{N-\eta+1}{N+2}$$

mixed $(1,2)$**, class C7,** $(1;2;12,23)$ $\quad L_+ = \{3\}$, $L_- = \{2\}$, $L_0 = \{1\}$.

$$A_\emptyset = \frac{(N+1)(\eta-1)(N-\eta+1)}{2N(N+2)}$$
$$A_{\{1\}} = \frac{(N+1)(\eta+1)(N-\eta+1)}{2N(N+2)}$$

$$\Gamma_\emptyset = \frac{(N+1)(\eta-1)(N-\eta+1)}{2N(N+2)}$$
$$\Gamma_{\{1\}} = \frac{(N+1)(N-\eta+1)}{N(N+2)}$$

mixed $(1,2)$**, class C8,** $(1;2;13,13)$ $\quad L_+ = \{1\}$, $L_- = \{2\}$, $L_0 = \{3\}$.

$$A_\emptyset = \frac{(\eta-1)(N-\eta+1)}{2N(N+2)}$$
$$A_{\{3\}} = \frac{(N-\eta+1)(2N+\eta+1)}{2N(N+2)}$$

$$\Gamma_\emptyset = \frac{(\eta-1)(N-\eta+1)}{2N(N+2)}$$
$$\Gamma_{\{3\}} = \frac{(N+1)(N-\eta+1)}{N(N+2)}$$

mixed $(1,2)$**, class C9,** $(1;1;23,24)$ $\quad L_+ = \{4\}$, $L_- = \{3\}$, $L_0 = \{1,2\}$.

$$A_\emptyset = -\frac{(N+1)(\eta-1)(N-\eta+1)}{2N(N-1)(N+2)}$$
$$A_{\{1\}} = -\frac{(N+1)(\eta+1)(N-\eta+1)}{2N(N-1)(N+2)}$$
$$A_{\{2\}} = -\frac{(N-\eta+1)(N\eta+3N+\eta-1)}{2N(N-1)(N+2)}$$
$$A_{\{1,2\}} = \frac{2N^3-N^2\eta+N^2+N\eta^2+3N\eta-2N+\eta^2-1}{2N(N-1)(N+2)}$$

$$\Gamma_{\emptyset} = -\frac{(N+1)(\eta-1)(N-\eta+1)}{2N(N-1)(N+2)}$$
$$\Gamma_{\{1\}} = -\frac{(N+1)(N-\eta+1)}{N(N-1)(N+2)}$$
$$\Gamma_{\{2\}} = -\frac{2(N-\eta+1)}{(N-1)(N+2)}$$
$$\Gamma_{\{1,2\}} = \frac{N+1}{N-1}$$

mixed $(1,2)$**, class C10,** $(1;2;13,24)$ $\quad L_+ = \{4\}$, $L_- = \{3\}$, $L_0 = \{1,2\}$.

$$A_{\emptyset} = \frac{(N+1)(\eta-1)(N-\eta+1)}{2(N-1)(N+2)}$$
$$A_{\{1\}} = \frac{(N+1)(\eta+1)(N-\eta+1)}{2(N-1)(N+2)}$$
$$A_{\{2\}} = \frac{(N+1)(\eta+1)(N-\eta+1)}{2(N-1)(N+2)}$$
$$A_{\{1,2\}} = \frac{(N+1)(\eta+1)(N-\eta-1)}{2(N-1)(N+2)}$$

$$\Gamma_{\emptyset} = \frac{(N+1)(\eta-1)(N-\eta+1)}{2(N-1)(N+2)}$$
$$\Gamma_{\{1\}} = \frac{(N+1)(N-\eta+1)}{(N-1)(N+2)}$$
$$\Gamma_{\{2\}} = \frac{(N+1)(N-\eta+1)}{(N-1)(N+2)}$$
$$\Gamma_{\{1,2\}} = -\frac{N+1}{N-1}$$

mixed $(1,2)$**, class C11,** $(1;2;13,34)$ $\quad L_+ = \{4\}$, $L_- = \{2\}$, $L_0 = \{1,3\}$.

$$A_{\emptyset} = -\frac{(N+1)(\eta-1)(N-\eta+1)}{2N(N-1)(N+2)}$$
$$A_{\{1\}} = -\frac{(N+1)(\eta+1)(N-\eta+1)}{2N(N-1)(N+2)}$$
$$A_{\{3\}} = -\frac{(N+1)(N-\eta+1)(2N+\eta-1)}{2N(N-1)(N+2)}$$
$$A_{\{1,3\}} = \frac{(N+1)(\eta+1)(N+\eta-1)}{2N(N-1)(N+2)}$$

$$\Gamma_{\emptyset} = -\frac{(N+1)(\eta-1)(N-\eta+1)}{2N(N-1)(N+2)}$$
$$\Gamma_{\{1\}} = -\frac{(N+1)(N-\eta+1)}{N(N-1)(N+2)}$$

$$\Gamma_{\{3\}} = -\frac{(N+1)(N-\eta+1)}{(N-1)(N+2)}$$
$$\Gamma_{\{1,3\}} = \frac{N+1}{N-1}$$

mixed $(1,2)$**, class C12,** $(1;2;34,34)$ $\quad L_+ = \{1\}$, $L_- = \{2\}$, $L_0 = \{3,4\}$.

$$A_{\emptyset} = -\frac{(\eta-1)(N-\eta+1)}{N(N-1)(N+2)}$$
$$A_{\{3\}} = -\frac{(N+\eta)(N-\eta+1)}{N(N-1)(N+2)}$$
$$A_{\{4\}} = -\frac{(N+\eta)(N-\eta+1)}{N(N-1)(N+2)}$$
$$A_{\{3,4\}} = \frac{N^3+N^2+N\eta-N+\eta^2-1}{N(N-1)(N+2)}$$

$$\Gamma_{\emptyset} = -\frac{(\eta-1)(N-\eta+1)}{N(N-1)(N+2)}$$
$$\Gamma_{\{3\}} = -\frac{(N+1)(N-\eta+1)}{N(N-1)(N+2)}$$
$$\Gamma_{\{4\}} = -\frac{(N+1)(N-\eta+1)}{N(N-1)(N+2)}$$
$$\Gamma_{\{3,4\}} = \frac{N+1}{N-1}$$

mixed $(1,2)$**, class C13,** $(1;1;12,34)$ $\quad L_+ = \{3,4\}$, $L_- = \{1,2\}$, $L_0 = \emptyset$.

$$A_{\emptyset} = \frac{N-\eta+1}{N+2}$$

$$\Gamma_{\emptyset} = \frac{N-\eta+1}{N+2}$$

mixed $(1,2)$**, class C14,** $(1;2;13,14)$ $\quad L_+ = \{1,4\}$, $L_- = \{2,3\}$, $L_0 = \emptyset$.

$$A_{\emptyset} = \frac{(N+1)(N-\eta+1)}{N(N+2)}$$

$$\Gamma_{\emptyset} = \frac{(N+1)(N-\eta+1)}{N(N+2)}$$

mixed $(1,2)$**, class C15,** $(1;1;23,45)$ $\quad L_+ = \{4,5\}$, $L_- = \{2,3\}$, $L_0 = \{1\}$.

$$A_{\emptyset} = -\frac{2(N-\eta+1)}{(N-1)(N+2)}$$

$$A_{\{1\}} = \frac{N^2 + N + 2\eta}{(N-1)(N+2)}$$

$$\Gamma_{\emptyset} = -\frac{2(N-\eta+1)}{(N-1)(N+2)}$$
$$\Gamma_{\{1\}} = \frac{N+1}{N-1}$$

mixed $(1,2)$**, class C16,** $(1;2;13,45)$ $\quad L_+ = \{4,5\}$, $L_- = \{2,3\}$, $L_0 = \{1\}$.

$$A_{\emptyset} = \frac{(N+1)(N-\eta+1)}{(N-1)(N+2)}$$
$$A_{\{1\}} = -\frac{(N+1)(\eta+1)}{(N-1)(N+2)}$$

$$\Gamma_{\emptyset} = \frac{(N+1)(N-\eta+1)}{(N-1)(N+2)}$$
$$\Gamma_{\{1\}} = -\frac{N+1}{N-1}$$

mixed $(1,2)$**, class C17,** $(1;2;34,35)$ $\quad L_+ = \{1,5\}$, $L_- = \{2,4\}$, $L_0 = \{3\}$.

$$A_{\emptyset} = -\frac{(N+1)(N-\eta+1)}{N(N-1)(N+2)}$$
$$A_{\{3\}} = \frac{(N+1)\left(N^2+N+\eta-1\right)}{N(N-1)(N+2)}$$

$$\Gamma_{\emptyset} = -\frac{(N+1)(N-\eta+1)}{N(N-1)(N+2)}$$
$$\Gamma_{\{3\}} = \frac{N+1}{N-1}$$

mixed $(1,2)$**, class C18,** $(1;2;34,56)$ $\quad L_+ = \{1,5,6\}$, $L_- = \{2,3,4\}$, $L_0 = \emptyset$.

$$A_{\emptyset} = \frac{N+1}{N-1}$$

$$\Gamma_{\emptyset} = \frac{N+1}{N-1}$$